

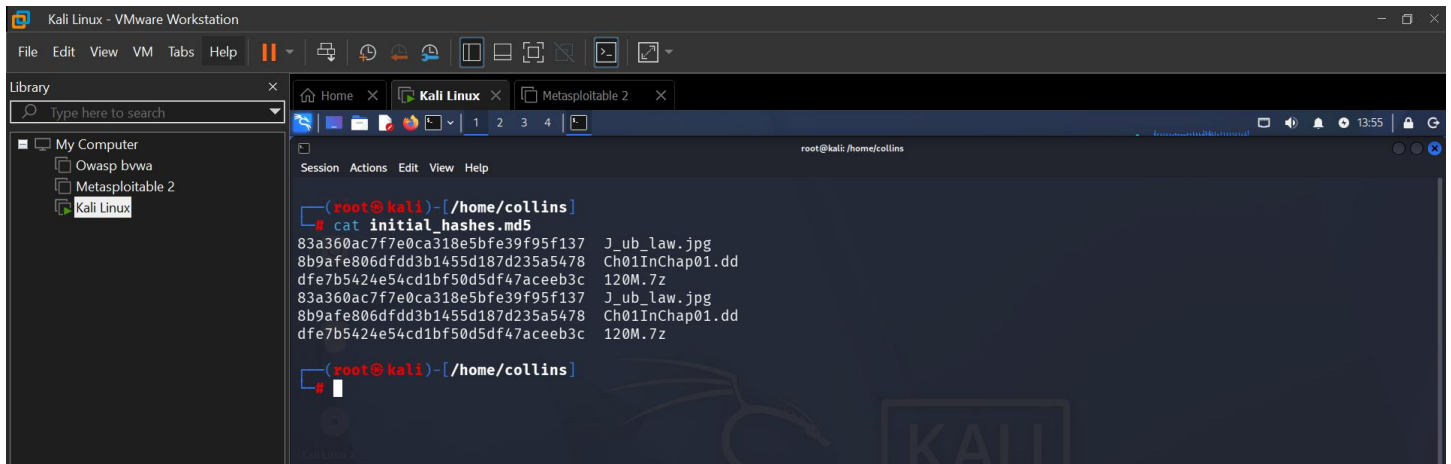
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COURSE CODE: SBT-DF202
LAB NAME: DATA CARVING WITH XXD, BINWALK AND SCALPEL
DATE: 04/09/2026
INSTRUCTOR: AMINU IDRIS

Laboratory instructions

Before beginning this lab, review the **Module 7 lessons and study material on Data Carving and File Recovery Techniques**. The lecture material is already a detailed walkthrough, so you are expected to understand file signatures, hexadecimal analysis, embedded-file recovery and carving concepts before executing commands.

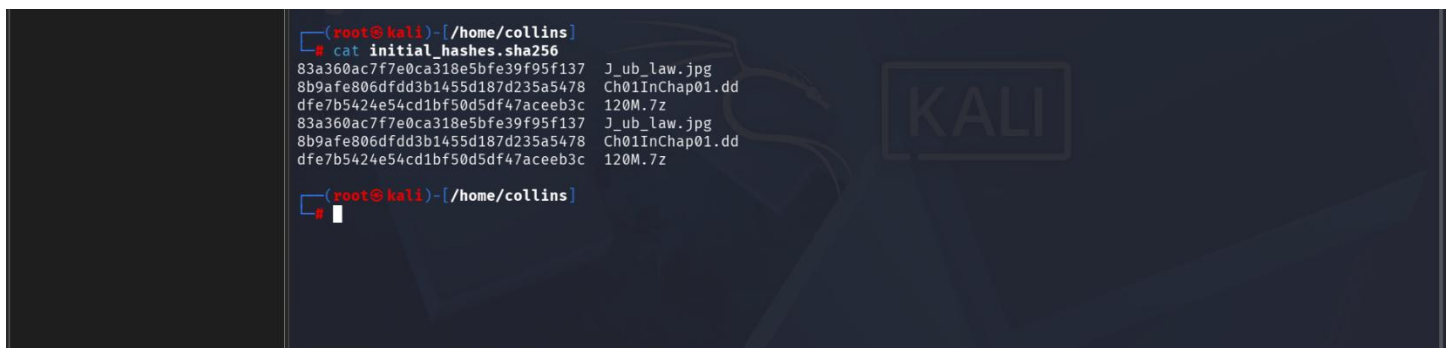
Create a dedicated working directory and preserve all downloaded evidence files. Record MD5 and SHA-256 hashes before beginning analysis.

ANSWER:



```
(root@kali)-[/home/collins]
# cat initial_hashes.md5
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
8b9afe806dfd3b1455d187d235a5478 Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
8b9afe806dfd3b1455d187d235a5478 Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(root@kali)-[/home/collins]
#
```



```
(root@kali)-[/home/collins]
# cat initial_hashes.sha256
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
8b9afe806dfd3b1455d187d235a5478 Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
8b9afe806dfd3b1455d187d235a5478 Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(root@kali)-[/home/collins]
#
```

Required Lab Resources

Download the authorised training evidence below:

- [Ch01InChap01.dd — Forensic DD Image](#)

- J_ub_law.jpg — JPEG Sample for Hex and Signature Analysis
- 120M.7z — USB Image for Scalpel File Carving

These are the exact evidence resources identified in the lab guide.

Part A - File Signatures and Hex Analysis

Use xxd to inspect J_ub_law.jpg, identify the JPEG header and footer signatures, create a plain hexadecimal dump, and reconstruct the image using reverse-hexdump functionality.

Verify the reconstructed image by comparing MD5 and SHA-256 hashes with the original.

ANSWER:

```
(root@kali)-[/home/collins]
# xxd -l 16 J_ub_law.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 ... Exif..II*.

(root@kali)-[/home/collins]
# xxd J_ub_law.jpg | tail -n 2
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8...o..Z.T.e
0019d1f0: d351 9ebf ffd9                .Q....

(root@kali)-[/home/collins]
#
```

```
(root@kali)-[/home/collins]
# xxd -p J_ub_law.jpg > J_ub_law.hex

(root@kali)-[/home/collins]
# xxd -r -p J_ub_law.hex reconstructed_J_ub_law.jpg

(root@kali)-[/home/collins]
#
```

```
(root@kali)-[/home/collins]
# md5sum J_ub_law.jpg reconstructed_J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 reconstructed_J_ub_law.jpg

(root@kali)-[/home/collins]
# sha256sum J_ub_law.jpg reconstructed_J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 reconstructed_J_ub_law.jpg

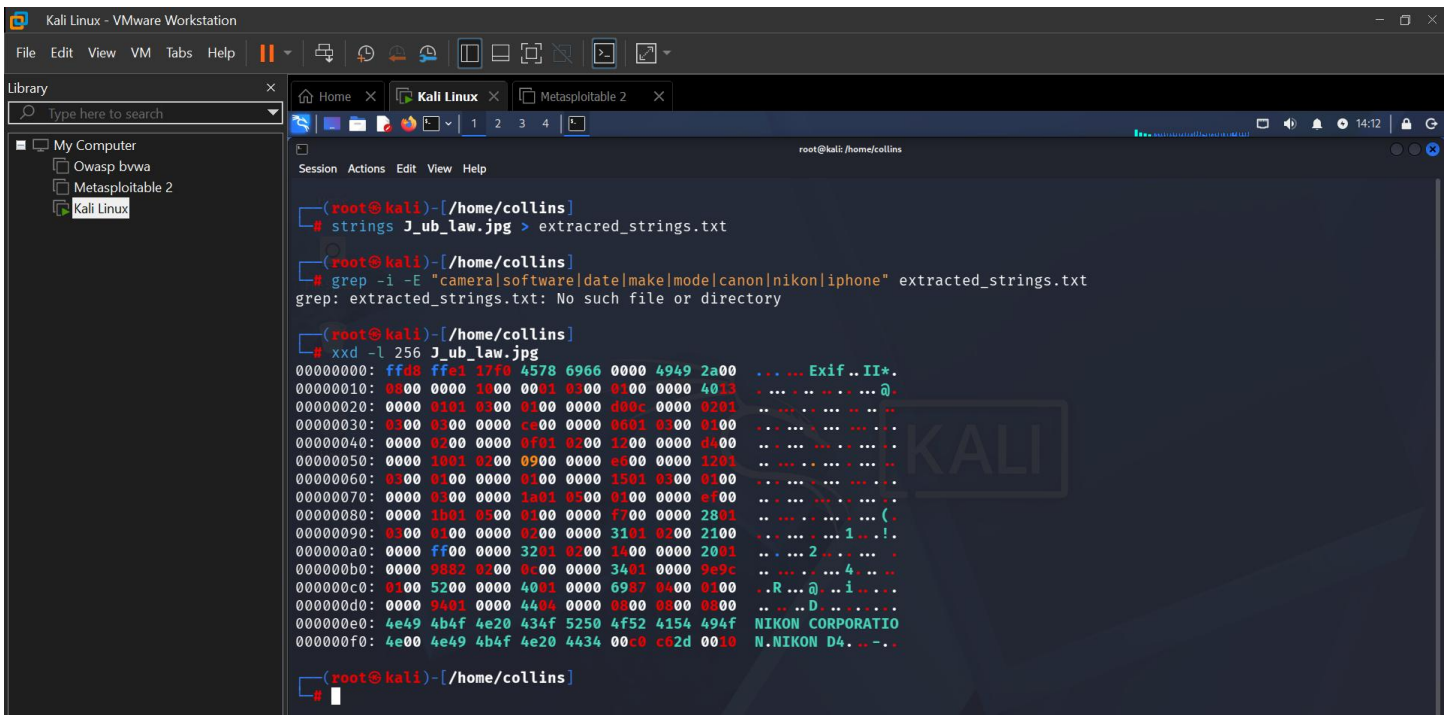
(root@kali)-[/home/collins]
#
```

Part B - Examine Embedded Metadata-Like Content

Use xxd and strings to inspect the JPEG for visible camera, software, name, date or other metadata-related strings.

Record only information actually found. If a searched item is not present, state **“not found”** rather than guessing.

ANSWER:



```
(root@kali)-[/home/collins]
# strings J_ub_law.jpg > extracted_strings.txt

(root@kali)-[/home/collins]
# grep -i -E "camera|software|date|make|model|canon|nikon|iphone" extracted_strings.txt
grep: extracted_strings.txt: No such file or directory

(root@kali)-[/home/collins]
# xxd -l 256 J_ub_law.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ... ..Exif..II*.
00000010: 0800 0000 1800 0081 0300 0100 0000 4013  .. ... ..@.
00000020: 0000 8101 0500 0100 0000 d80c 0000 0201  .. ... ..
00000030: 0300 0300 0000 c000 0000 0602 0300 0100  .. ... ..
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  .. ... ..
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  .. ... ..
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  .. ... ..
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  .. ... ..
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  .. ... ..
00000090: 0300 0100 0000 0700 0000 3101 0200 2101  .. ... ..
000000a0: 0000 ff00 0000 3201 0200 1a00 0000 2001  .. ... ..
000000b0: 0000 9802 0200 0c00 0000 3401 0000 9e9c  .. ... ..
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100  ..R...@...i...
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800  .. ...D...i...
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f  NIKON CORPORATION
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010  N.NIKON D4...-
```

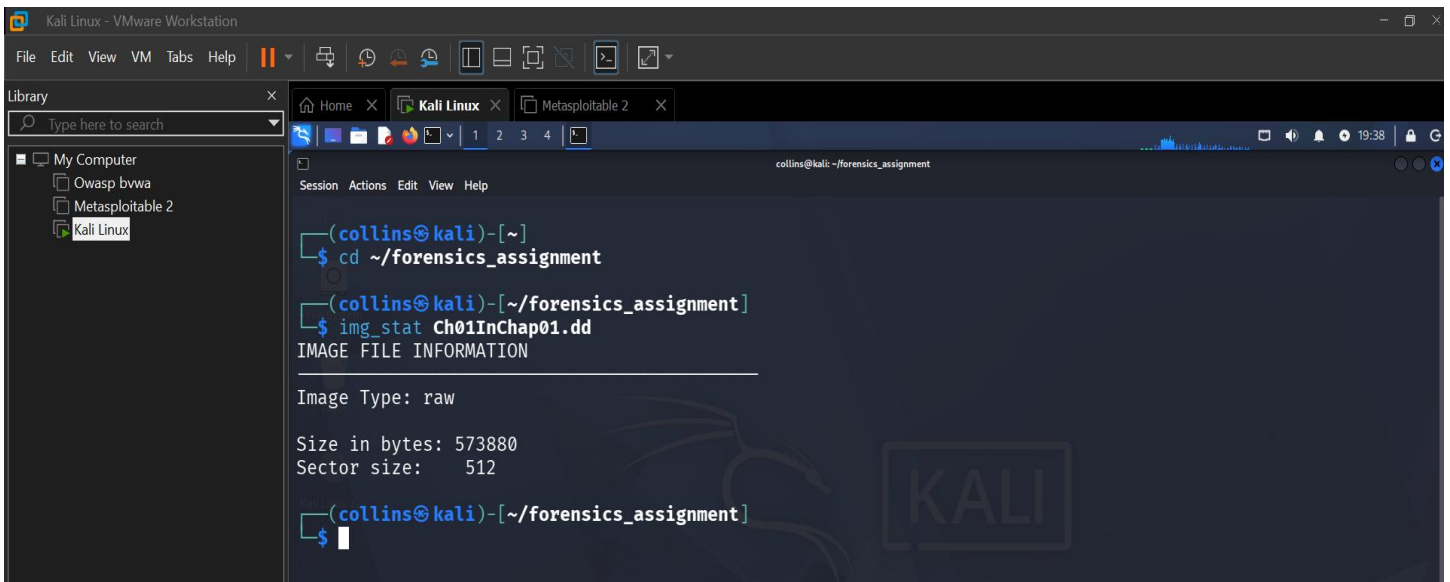
Part C - Direct Data-Unit Examination with The Sleuth Kit

Use tools such as:

- img_stat
- mmls
- fsstat
- fls
- icat
- blkcat

Identify actual inode and sector/block values from your own command output. Do not copy placeholder values from examples.

ANSWER:



```
(collins@kali)-[~]
$ cd ~/forensics_assignment

(collins@kali)-[~/forensics_assignment]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 573880
Sector size: 512

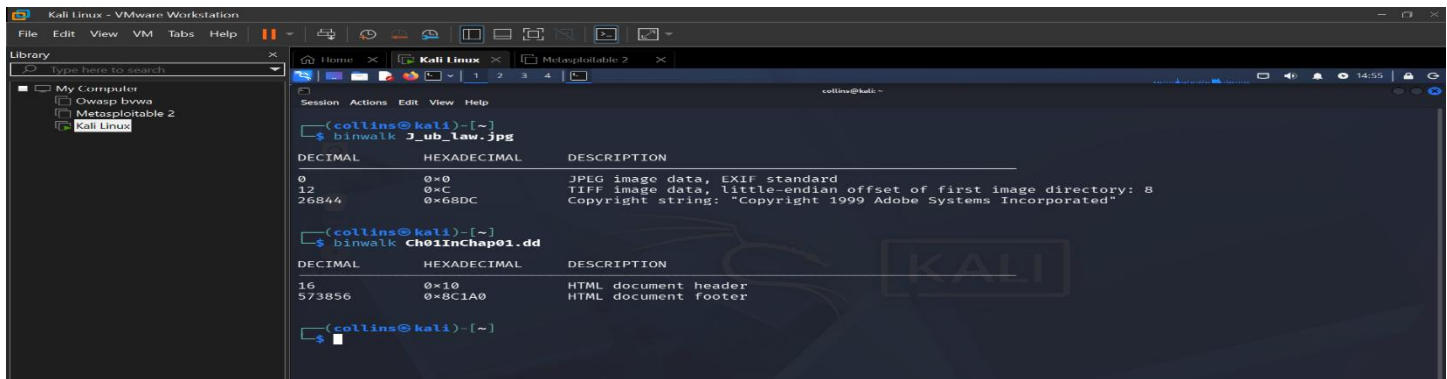
(collins@kali)-[~/forensics_assignment]
$
```

Part D - Embedded File Discovery with Binwalk

Use binwalk to identify embedded file structures or signatures.

If the instructor-provided File_carving.docx is available, perform the demonstrated manual and automatic extraction workflow. If it is not available, document this limitation and complete the Binwalk practice using the other supplied lab evidence.

ANSWER:



```
(collins@kali)-[~]
$ binwalk ub_law.jpg

DECIMAL      HEXADECFMAL    DESCRIPTION
-----
0            0x0            JPEG image data, EXIF standard
12           0xC            TIFF image data, little-endian offset of first image directory: 8
26844        0x68DC         Copyright string: "Copyright 1999 Adobe Systems Incorporated"

(collins@kali)-[~]
$ binwalk Ch01InChap01.dd

DECIMAL      HEXADECFMAL    DESCRIPTION
-----
16           0x10           HTML document header
573856       0x8C1A0        HTML document footer

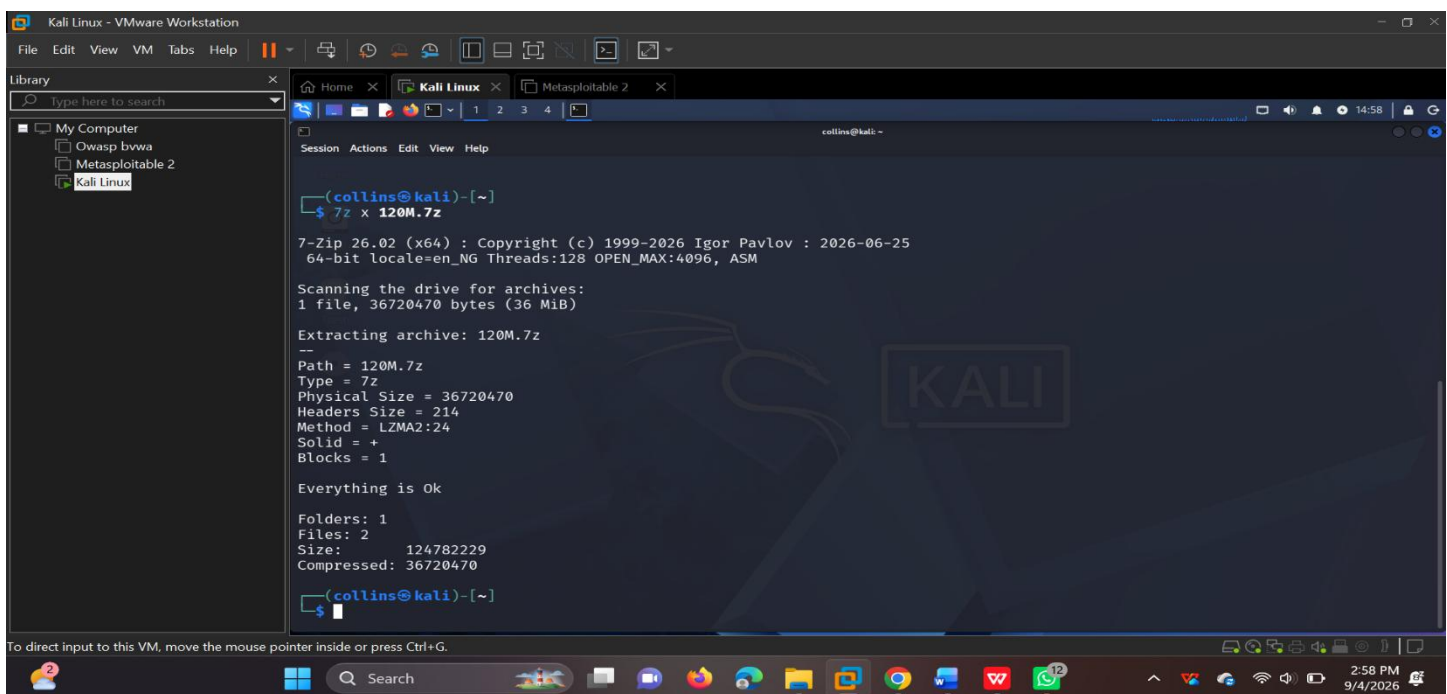
(collins@kali)-[~]
$
```

Part E - Prepare the USB Image

Extract 120M.7z, identify the resulting USB image, calculate hashes, inspect the image using TSK, and confirm the appropriate partition offset using mmls.

The source lab refers to an offset of 128, but students must confirm the partition structure rather than blindly copying the value.

ANSWER:



```
(collins@kali)-[~]
$ 7z x 120M.7z

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_NG Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size: 124782229
Compressed: 36720470

(collins@kali)-[~]
$
```



```
(collins@kali)-[~]
$ md5sum 120M.7z
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(collins@kali)-[~]
$ sha256sum 120M.7z
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z

(collins@kali)-[~]
$
```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

Part F - Carve Deleted Files with Scalpel

Configure Scalpel to enable the required file types, particularly JPEG/JPG, and run the carving operation against the forensic USB image.

Review:

- Recovered files
- File types
- Hashes
- audit.txt
- At least two recovered JPEG images where available

```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Owasp bwba
  Metasploitable 2
  Kali Linux

(collins@kali)-[~/forensics_assignment]
$ sudo rm -rf carved_output/

(collins@kali)-[~/forensics_assignment]
$ sudo scalpel -c /etc/scalpel/scalpel.conf -o carved_output_new/ Ch01InChap01.dd
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/collins/forensics_assignment/Ch01InChap01.dd"

Image file pass 1/2.
Ch01InChap01.dd: 100.0% |*****| 560.4 KB 00:00 ETA
Allocating work queues ...
Work queues allocation complete. Building carve lists ...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" -> 0 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" -> 0 files
pdf with header "\x25\x50\x44\x46" and footer "\x25\x45\x4f\x46\x0d" -> 0 files
pdf with header "\x25\x50\x44\x46" and footer "\x25\x45\x4f\x46\x0a" -> 0 files
Carving files from image.
Image file pass 2/2.
Ch01InChap01.dd: 100.0% |*****| 560.4 KB 00:00 ETA
Processing of image file complete. Cleaning up ...
Done.
Scalpel is done, files carved = 0, elapsed = 0 seconds.

(collins@kali)-[~/forensics_assignment]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ANSWER: